



National Institute of
Diabetes and Digestive
and Kidney Diseases

Type 1 Diabetes

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What is type 1 diabetes?

Type 1 diabetes is a disease that occurs when your body makes little or no [insulin](#). Insulin is a [hormone](#) that helps [blood glucose](#), also called blood sugar, get into the body's cells to be used as energy. When your body doesn't have enough insulin, glucose in your blood can't get into the cells. As a result, your cells lack energy, your blood glucose level goes up, and you develop [diabetes](#).

A problem with the body's [immune system](#) causes type 1 diabetes. Your immune system normally fights infections. Type 1 diabetes develops when the body's immune system destroys the cells in the [pancreas](#) that make insulin. When the immune system attacks healthy cells in your body, it's called autoimmunity. Type 1 diabetes is an autoimmune disease.

Most people with type 1 diabetes need to take insulin every day to manage their blood glucose level and stay alive. You may be able to prevent or delay [health problems](#) that can result from type 1 diabetes by learning to manage your blood glucose level, consuming healthy foods and drinks, and getting regular physical activity.



People with type 1 diabetes need to manage the disease to prevent health problems.

How common is type 1 diabetes?

In 2021, about 1.7 million U.S. adults ages 20 years or older—or 5.7% of adults with diagnosed diabetes—had type 1 diabetes and were taking insulin.¹ Overall, 5% to 10% of people with diabetes have type 1 diabetes.²

The number of young people with type 1 diabetes has been increasing over the last two decades. In 2021, about 304,000 children and teens younger than age 20 had type 1 diabetes.¹

Who is more likely to have type 1 diabetes?

Type 1 diabetes occurs more commonly in children and young adults, but it can develop at any age. Having a parent or sibling with type 1 diabetes increases your chance of developing the disease. People diagnosed with type 1 diabetes have the disease for the rest of their lives. Some people with type 1 diabetes can get a [pancreatic islet or pancreas transplant](#) and make insulin again. However, these transplants don't cure type 1 diabetes. People still need to take medicines called [immunosuppressants](#) to prevent the immune system from attacking and destroying the new islets or pancreas.

Type 1 diabetes is increasing among youth from certain racial and ethnic groups. However, new cases of type 1 diabetes are most common among White youth. Adults with diabetes who are younger than age 35 and have a healthy body weight are more likely to have type 1 diabetes than type 2 diabetes.²

What health problems can people with type 1 diabetes develop?

People with type 1 diabetes can develop the same [diabetes health problems](#) as people with other types of diabetes. High blood glucose levels from diabetes can damage many parts of your body, including your

- heart
- kidneys
- nerves
- eyes
- feet
- teeth

People with diabetes may also have [high blood pressure](#) [NIH](#) or [cholesterol](#) [NIH](#) problems. Diabetes is also linked to some types of [cancer](#) [NIH](#).

If you have type 1 diabetes, [managing your diabetes](#) and having a [healthy lifestyle](#) may lower your chances of developing health problems.

Diabetic ketoacidosis

People with type 1 diabetes also have a higher risk of developing a condition called [diabetic ketoacidosis \(DKA\)](#) [NIH](#). When you have type 1 diabetes, your body doesn't have enough insulin to use blood glucose for energy. Your body will begin to use [fat](#) as a source of energy instead. This process produces substances called [ketones](#). High levels of ketones can cause DKA. People have a higher risk of developing DKA when they have an infection or when they're under [stress](#) [NIH](#). DKA is a medical emergency that needs to be treated right away.

Hypoglycemia unawareness

Some people develop a serious condition called hypoglycemia unawareness. People with this condition don't feel their blood glucose level dropping, so they aren't aware they should treat low blood glucose, also called [hypoglycemia](#).

If you have type 1 diabetes, you'll need to carefully manage the amount of insulin you take every day to keep your blood glucose level in a healthy range. If you take too much insulin, you may develop low blood glucose.

Hypoglycemia unawareness is a medical emergency that needs to be treated right away. That's because hypoglycemia unawareness can cause problems with thinking and other brain functions and, if not treated, may lead to a [coma](#).

Mental health conditions

The stress of managing type 1 diabetes can make some people more likely to develop [sleep problems](#) [NIH](#), [anxiety](#) [NIH](#), or [depression](#) [NIH](#). You may find it helpful to [speak with a counselor](#) [NIH](#) or your primary health care professional about your mental health. Your primary health care professional may be a doctor, physician assistant, or nurse practitioner.

Other health problems

People with type 1 diabetes may also have a higher risk of other health problems, including

- eating disorders, such as [anorexia](#), [bulimia](#), and [binge eating disorder](#)
- other [autoimmune diseases](#) [NIH](#)
- hip fractures in older adults due to [osteoporosis](#) [NIH](#)

If you're [pregnant and have diabetes](#), changes to your hormones may affect how your body uses insulin. Ask your health care professional if you should adjust your medicines during pregnancy to prevent health problems for you and your baby.

What are the symptoms of type 1 diabetes?

Symptoms of type 1 diabetes may include

- increased urination
- feeling very thirsty
- feeling very hungry, even after you've eaten
- blurred vision
- fatigue
- sores that don't heal
- frequent infections, such as [urinary tract infections](#), [skin infections](#) [NIH](#), or [yeast infections](#) [NIH](#)
- unexplained weight loss

Symptoms of type 1 diabetes usually develop quickly, over a few days or weeks. Children with type 1 diabetes typically develop symptoms over a short period of time. Common symptoms in children include

- having to urinate more often
- eating more but losing weight
- feeling very thirsty



Feeling thirsty is a common symptom of type 1 diabetes in children.

These symptoms are less common in adults with type 1 diabetes. For some adults, symptoms of type 1 diabetes may develop slowly and seem more like [type 2 diabetes](#). Some adults who develop diabetes may need special tests to find out what type of diabetes they have.

The first symptoms of type 1 diabetes can be symptoms of [DKA](#), especially in children. Symptoms of DKA include

- feeling very tired
- having trouble breathing
- having fruity-smelling breath
- fainting from [dehydration](#)
- having pain in your [abdomen](#), [nausea](#), or [vomiting](#)

What causes type 1 diabetes?

Type 1 diabetes develops when the body's immune system destroys the cells in the pancreas that make insulin. These cells are called beta cells. Genes and factors in the environment—the place where people live, play, work, study, and gather—may trigger the immune system to destroy beta cells in type 1 diabetes.

Studies such as [The Environmental Determinants of Diabetes in the Young \(TEDDY\)](#)  are trying to learn more about the causes of type 1 diabetes. [Type 1 Diabetes TrialNet](#)  and other groups are working to find treatments that may prevent or slow the development of type 1 diabetes in people who are at risk for getting the disease.

How do health care professionals diagnose type 1 diabetes?

Your health care professional can diagnose diabetes by reviewing your [symptoms](#), asking about your medical history, and using diabetes tests. Your health care professional may also have you take [diabetes tests](#) to find out if your diabetes is type 1, type 2, or another type of diabetes. Your treatment will depend on the type of diabetes you have.

Health care professionals often test for type 1 diabetes in people who have common diabetes symptoms. Since type 1 diabetes can run in families, your health care professional may test you for type 1 diabetes if you have no symptoms but have a parent or sibling with type 1 diabetes.

Blood glucose tests

Health care professionals can diagnose diabetes using blood tests that measure your blood glucose level to see if it's higher than normal. Health care professionals may use

- the [random plasma glucose test](#) to diagnose diabetes in people who have diabetes symptoms
- the [fasting plasma glucose test](#)
- the [A1C test](#)

The random plasma glucose test and fasting plasma glucose test measure your blood glucose level at the time your blood is drawn. The A1C test shows your average blood glucose level over the past 3 months.

Although these blood glucose tests can help your health care professional tell if you have diabetes, the tests don't show the type of diabetes you have.

Autoantibodies test

To find out if your diabetes is type 1, your health care professional may test your blood for certain autoantibodies.

Autoantibodies and type 1 diabetes

Autoantibodies are proteins made by your immune system that attack healthy tissues and cells by mistake. If you have certain autoantibodies in your blood, you may have type 1 diabetes.

Autoantibodies and latent autoimmune diabetes in adults

Some adults with diabetes have autoantibodies in their blood but develop symptoms slower than most people with type 1 diabetes. In this type of diabetes, your immune system damages your pancreas more slowly. You may not need to take insulin for several months after being diagnosed. This type of diabetes is sometimes called latent autoimmune diabetes in adults (LADA).

Autoantibodies and family members of people with type 1 diabetes

If you have a parent or sibling with type 1 diabetes, your health care professional may suggest a test for autoantibodies, even if you don't have any symptoms. The test may show if you have a higher risk of developing symptoms of type 1 diabetes and if you should be treated to delay the disease. Talk with your health care professional about the symptoms of diabetes and ask if they want you to get more testing in the future.

TrialNet is a research network that offers [autoantibody testing](#)  to some family members of people with type 1 diabetes. Some people at risk of developing symptoms of type 1 diabetes may be able to participate in TrialNet studies.

Other tests

If it's not clear what type of diabetes you have, your health care professional may ask you to do

- a [C-peptide test](#)  to see how much insulin your pancreas makes
- [genetic testing](#)  to rule out less common types of diabetes, such as [monogenic diabetes](#)

How can I manage type 1 diabetes?

If you have type 1 diabetes, you'll need to take insulin to manage your blood glucose level. Using devices to check your blood glucose level and inject insulin may help keep your blood glucose level in a healthy range. Your health care professional may also prescribe other diabetes medicines.

Managing your type 1 diabetes may help prevent or delay diabetes health problems.

Having a healthy lifestyle can help manage your type 1 diabetes and prevent other health problems. With the help of your health care team, you can create a diabetes care plan that works for you.

Diabetes medicines, technologies, and office visits can be expensive. Tell your health care professional or health care team if you're having trouble affording diabetes care. A health care professional, diabetes educator, or social worker may be able to tell you about resources that provide [financial help for diabetes care](#).

Insulin and other diabetes medicines

If you have type 1 diabetes, you need to take insulin to stay alive. Different [types of insulin](#) are available. Some types start to work quickly while others start to work more slowly. The effects of each type of insulin last for varying lengths of time. You may need to take more than one type of insulin. You can take insulin in a few ways, including

- [needle and syringe](#)
- [insulin pen](#)
- [insulin pump](#)

If you take insulin, you need to check your blood glucose level regularly. You should match your insulin dose with your food, physical activity, and current blood glucose level. If you don't, your blood glucose level may drop too low, which is called hypoglycemia. Hypoglycemia is a medical emergency that needs to be treated right away.

Some people have trouble managing their blood glucose level with just insulin. Your health care professional may recommend other medicines to take with insulin to help manage your blood glucose level.

Diabetes devices and technology

Diabetes devices and technology can help people with type 1 diabetes take insulin and check their blood glucose level.

Insulin pump

An insulin pump is a small, portable machine that gives you frequent, small doses of insulin using a plastic tube under the skin. The pump is refilled with insulin every 2 to 3 days. Insulin pumps can be programmed to deliver insulin directly into your body 24 hours a day or at certain times.

Blood glucose meter

People with diabetes can check their blood glucose level with a blood glucose meter. After drawing a small amount of blood from your finger, you can use the meter to measure how much glucose is in your blood at that moment.

Continuous glucose monitor

A [continuous glucose monitor \(CGM\)](#) is a device that automatically tracks your blood glucose level throughout the day and night. Researchers have found that using a CGM at home may help you manage your blood glucose level and reduce your risk of developing health problems from diabetes.

CGMs use sensors placed under your skin that estimate your blood glucose level. You can see information about your blood glucose level at any time in a software program stored on a smartphone, on an insulin pump, or on a separate device called a receiver. These results may help you and your health care team make

changes to your meal plan, physical activity level, or diabetes medicines. Many CGM systems have alarms and warnings that let you know if your blood glucose level is too low or too high. For safety, it's important to act quickly if a CGM alarm sounds.



Diabetes technologies such as CGMs can help you keep your blood glucose level in a healthy range.

Artificial pancreas

An **artificial pancreas** is a system that mimics how a healthy pancreas controls blood glucose levels in the body. The system is made up of three parts. A CGM automatically estimates your blood glucose level and sends the information, without using wires, to a software program on your smartphone or on an insulin pump. The software program calculates how much insulin your body needs, and the insulin pump automatically delivers that amount of insulin to your body.

Researchers have found that artificial pancreas systems may control blood glucose levels better than other methods in adults and children as young as age 2 years. This is especially true at night, when managing blood glucose levels can be hard for many people with type 1 diabetes.

Ask your health care professional which diabetes devices will work for you.

Pancreatic islet and pancreas transplant

Pancreatic islet transplant

A [pancreatic islet transplant](#) is used to treat type 1 diabetes in people who struggle to manage their blood glucose levels. Pancreatic islets are clusters of cells in the pancreas that make insulin. In type 1 diabetes, the body's immune system attacks these cells. A pancreatic islet transplant replaces destroyed islets with new islets from a deceased donor. The new islets make and release insulin. After a pancreatic islet transplant, you'll take immunosuppressants to prevent the immune system from attacking and destroying the new islets.

Pancreas transplant

Health care professionals may consider a [pancreas transplant](#) [NIH](#) for people with type 1 diabetes who need a [kidney transplant](#) or who haven't been able to manage their blood glucose levels in any other way. A pancreas transplant may restore normal blood glucose levels and prevent other diabetes-related health problems. After a pancreas transplant, you'll take immunosuppressants to prevent the immune system from attacking and destroying the new pancreas.

Can I delay or prevent type 1 diabetes in me or in my family members who may be at risk?

[Teplizumab](#) [NIH](#) is a medicine that can delay type 1 diabetes symptoms from developing and delay the need for insulin injections. If you or a family member has two or more autoantibodies for type 1 diabetes and increasing blood glucose levels, your health care professional may recommend this medicine.

Through studies such as those conducted by TrialNet, researchers are working to identify more risk factors for type 1 diabetes and find new ways to delay and prevent the disease.

Clinical Trials for Type 1 Diabetes

NIDDK conducts and supports clinical trials in many diseases and conditions, including type 1 diabetes. The trials look to find new ways to prevent, detect, or treat disease and improve quality of life.

What are clinical trials for type 1 diabetes?

Clinical trials—and other types of [clinical studies](#) [NIH](#)—are part of medical research and involve people like you. When you volunteer to take part in a clinical study, you help doctors and researchers learn more about disease and improve health care for people in the future.

Researchers are studying many aspects of type 1 diabetes, such as

- improving the safety and success of pancreatic islet transplantation
- reducing the immune system response that destroys cells that produce insulin
- looking at insulin resistance in adolescents with type 1 diabetes

[Find out if clinical trials are right for you](#) [NIH](#) .

Watch a video of NIDDK Director Dr. Griffin P. Rodgers explaining the importance of participating in clinical trials.

Why Should I Join a Clinical Trial?

National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK)

From a US national research
authority



Watch on

What clinical trials for type 1 diabetes are looking for participants?

You can view a filtered list of clinical studies on type 1 diabetes that are federally funded, open, and recruiting at [ClinicalTrials.gov](#) . You can expand or narrow the list to include clinical studies from industry, universities, and individuals; however, the National Institutes of Health does not review these studies and cannot ensure they are safe for you. Always talk with your primary health care professional before you participate in a clinical study.

What have we learned about type 1 diabetes from NIDDK-funded research?

Research funded by NIDDK has improved the quality of life for people with type 1 diabetes by finding better ways to diagnose and treat the disease.

Delaying the onset of type 1 diabetes

NIDDK funded research on delaying the onset of type 1 diabetes, including the clinical trials network TrialNet. This research was instrumental in developing the first medicine to delay symptoms of type 1 diabetes, teplizumab.

Preventing health problems related to type 1 diabetes

NIDDK funded the landmark [Diabetes Control and Complications Trial \(DCCT\)](#). The study showed that participants with type 1 diabetes who kept their blood glucose levels close to normal greatly lowered their chances of developing eye, kidney, and nerve disease.

A follow-up study, the ongoing [Epidemiology of Diabetes Interventions and Complications \(EDIC\)](#)  study, has continued to follow DCCT participants since 1994. Findings from the DCCT and EDIC studies changed the way diabetes is treated worldwide, and they show that managing blood glucose levels can help people with diabetes lead longer and healthier lives.

Artificial pancreas

NIDDK continues to fund research into new technologies for managing blood glucose levels. A trial funded by NIDDK found that one artificial pancreas technology, a [device known as the bionic pancreas](#) [NIH](#) , improved type 1 diabetes management compared with standard insulin delivery methods. [The Pediatric Artificial Pancreas \(PEDAP\) Trial](#) [NIH](#)  found that artificial pancreas technology improved blood glucose level control in young children between ages 2 and 5 years with type 1 diabetes when compared with standard care methods.

References

[1] National diabetes statistics report. Centers for Disease Control and Prevention. Accessed October 18, 2024. www.cdc.gov/diabetes/php/data-research/index.html 

[2] American Diabetes Association Professional Practice Committee. 2. Diagnosis and classification of diabetes: standards of care in diabetes—2024. *Diabetes Care*. 2024;47(suppl 1):S20–S42. doi:10.2337/dc24-S002

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